

Case Study 40: Backend Development - Virtual Classroom

Context

Remote learning rooms. Learners will build the backend for a Virtual Classroom using Node.js, Express.js, MongoDB, Firebase, and [Socket.io](#).

Challenge

Develop the backend API for "ClassLive" where:

- Users can join classes and take quizzes.
- Share screens (mock) and track attendance.
- Real-time classroom updates via [Socket.io](#).
- Firebase push notifications for class schedules.

Requirements

- **Backend:** Node.js + Express.js REST API with modular routes and controllers.
- **Database:** MongoDB with Mongoose ODM for sessions, students, and quizzes.
- **Authentication:** JWT-based authentication + Firebase Auth.
- **WebSockets:** [Socket.io](#) for real-time classroom.
- **API Endpoints:**
 - Auth: POST /api/auth/register, POST /api/auth/login
 - Classes: GET /api/classes, GET /api/classes/:id, POST /api/classes (teacher), PUT /api/classes/:id
 - Sessions: POST /api/sessions, GET /api/sessions, GET /api/sessions/:id, PUT /api/sessions/:id
 - Attendance: POST /api/attendance, GET /api/attendance, GET /api/attendance/student/:id
 - Quizzes: GET /api/quizzes, GET /api/quizzes/:id, POST /api/quizzes (teacher)
 - Submissions: POST /api/submissions, GET /api/submissions, GET /api/submissions/student/:id
 - Notifications: POST /api/notifications/send
- **Middleware:** JWT authentication, role-based authorization, validation middleware.

Advanced Features (Optional)

- Recordings API.
- Breakouts API.

Deliverables

- REST API with all CRUD operations.
- WebSocket server for real-time classroom.
- API documentation (Swagger/Postman collection).
- GitHub repository with README and setup instructions.
- Bonus: Deployment on Render/Heroku.

Example User Flow

1. POST /api/sessions → Join room with [Socket.io](https://socket.io)
 2. POST /api/submissions → Take quiz
 3. POST /api/attendance → Attendance marked
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